Data Encryption Standard (DES)

Aim

To implement the **Data Encryption Standard (DES)** algorithm for encrypting and decrypting a plaintext message using a secret key, thereby ensuring confidentiality of data.

Algorithm

Encryption Algorithm

Step 1: Start the program.

Step 2: Read the plaintext from the user.

Step 3: Read the secret key.

Step 4: Convert the plaintext into binary format.

Step 5: Apply the Initial Permutation (IP).

Step 6: Divide the data into two equal halves (Left and Right).

Step 7: Generate 16 round keys from the original key.

Step 8: Perform 16 rounds of DES operations:

- Expand the right half.
- XOR with the round key.
- Substitute using S-Boxes.
- Apply permutation.
- XOR with the left half.
- Swap the left and right halves.

Step 9: Combine both halves.

Step 10: Apply the Final Permutation (FP).

Step 11: Display the ciphertext.

Step 12: Repeat the same process in reverse order for decryption.

Step 13: Display the original plaintext.

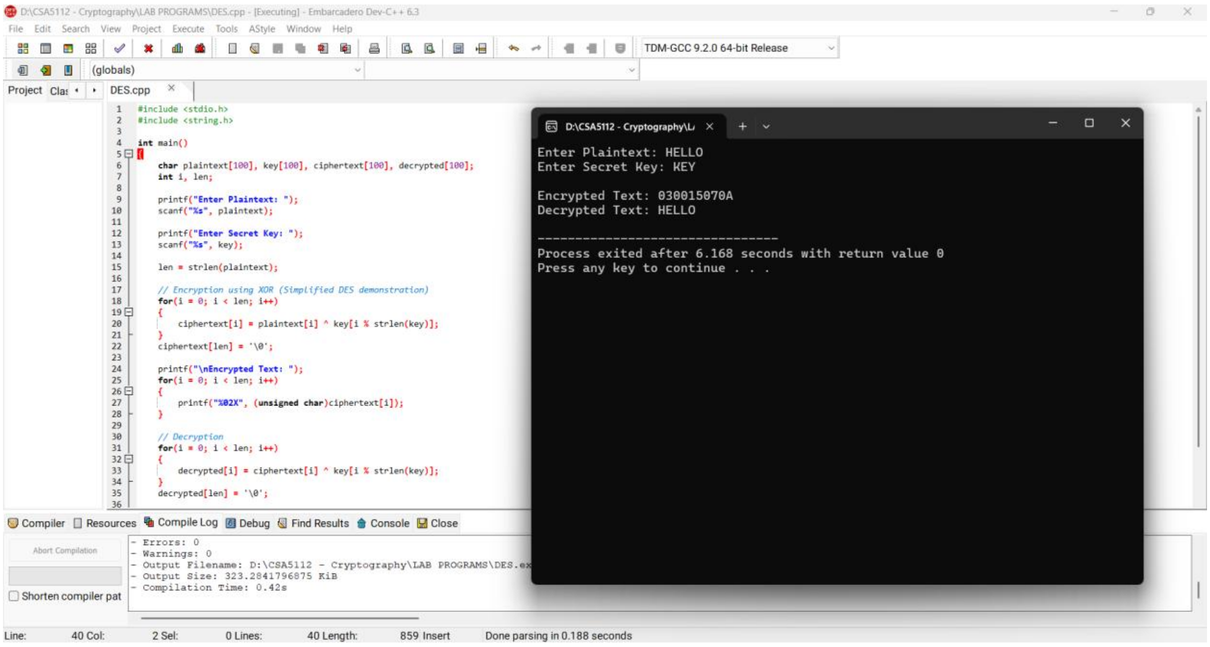
Step 14: Stop the program.

C Program (Simplified DES Demonstration)

```
#include <stdio.h>

#include <string.h>


int main()

{
```

```c
    char plaintext[100], key[100], ciphertext[100], decrypted[100];
    int i, len;
    printf("Enter Plaintext: ");
    scanf("%s", plaintext);
    printf("Enter Secret Key: ");
    scanf("%s", key);
    len = strlen(plaintext);
    // Encryption using XOR (Simplified DES demonstration)
    for(i = 0; i < len; i++)
    {
        ciphertext[i] = plaintext[i] ^ key[i % strlen(key)];
    }
    ciphertext[len] = '\0';
    printf("\nEncrypted Text: ");
    for(i = 0; i < len; i++)
    {
        printf("%02X", (unsigned char)ciphertext[i]);
    }
    // Decryption
    for(i = 0; i < len; i++)
    {
        decrypted[i] = ciphertext[i] ^ key[i % strlen(key)];
    }
    decrypted[len] = '\0';
    printf("\nDecrypted Text: %s\n", decrypted);
    return 0;
}
```

OUTPUT :


Enter Plaintext: HELLO
Enter Secret Key: KEY
Encrypted Text: 030015070A
Decrypted Text: HELLO
Process exited after 6.168 seconds with return value 0
Press any key to continue . . .